Newton-Raphson method

Suppose f(x) be twice continuously differentiable function on [a,b], i.e $f \in c^2[a,b]$. Let $f'(x) \neq 0$, let $\bar{x}$ be any point in [a,b] such that $f'(\bar{x}) \neq 0$, let p be the root of f(x), i.e $f(p) = 0$ and $|\bar{x} - p|$ be small enough, i.e $|\bar{x} - p| < \varepsilon$. Then (Taylor expansion for f(x) about $x = \bar{x}$)

$$f(x) \cong f(\bar{x}) + (x - \bar{x}) f'(\bar{x}) + \frac{(x - \bar{x})^2}{2!} f''(\eta), \eta \in (x, \bar{x})$$

since $f(p) = 0$ with $x = p$ gives

$$0 = f(p) \cong f(\bar{x}) + (p - \bar{x}) f'(\bar{x}) + \frac{(p - \bar{x})^2}{2!} f''(\eta), \eta \in (p, \bar{x})$$

$$\Rightarrow -(p - \bar{x}) f'(\bar{x}) = f(\bar{x})$$

$$\Rightarrow -(p - \bar{x}) = \frac{f(\bar{x})}{f'(\bar{x})} \Rightarrow p = \bar{x} - \frac{f(\bar{x})}{f'(\bar{x})}$$ Newton Raphson p is the approximate root for f(x), i.e $f(p) = 0$

$\bar{x}$ is any initial point $\in [a,b]$ and $\bar{x}$ closed to p.

The steps of solve Newton Raphson

$n = 1$

$p_1 = p_0 - \frac{f(p_0)}{f'(p_0)}, p_0 \in [a,b], p_0$ is initial point

$$p_2 = p_1 - \frac{f(p_1)}{f'(p_1)}$$

$$p_3 = p_2 - \frac{f(p_2)}{f'(p_2)}$$

$\vdots$

$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, n \geq 1$ Newton Raphson method

Stopping criterion

1. $|p_n - p_{n-1}| < \varepsilon$

2. $|f(p_n)| < \varepsilon$

3. $\frac{|p_n - p_{n-1}|}{|p_n|} < \varepsilon, p_n \neq 0$

Ex(1):- Find the root of the eq. $x^2 + 2.1x - 1 = 0$ by N.R with $x_0 = 0.5$

Solu. $f'(x) = 2x + 2.1$

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}, n \geq 1$$

$$x_1 = 0.5 - \frac{0.3}{3.1} = 0.4032$$

$$x_2 = 0.4032 - \frac{0.0093}{2.9064} = 0.4000$$

$$x_3 = 0.4000 - \frac{0}{2.9} = 0.4$$

$\therefore x_3 = 0.4$ is a root

Ex(2):- Use N.R to estimate $\sqrt{7}$ take $x_0 = 2.5$

Solu. Suppose $x = \sqrt{7} \Rightarrow x^2 = 7 \Rightarrow x^2 - 7 = 0$

$$f(x)=x^2-7 \Rightarrow f'(x)=2x$$

$$x_{n+1}=x_n-\frac{f(x)}{f'(x)}=x_n-\frac{x_n^2-7}{2x_n}=x_n-\left(\frac{x_n^2}{2x_n}-\frac{7}{2x_n}\right)=x_n-\left(\frac{x_n}{2}-\frac{7}{2x_n}\right)=x_n-\frac{x_n}{2}$$

$$-\frac{7}{2x_n}=\frac{1}{2}\left(x_n+\frac{7}{x_n}\right)$$

$$x_1=\frac{1}{2}\left(2.5+\frac{7}{2.5}\right)=2.65$$

$$x_2=\frac{1}{2}\left(2.65+\frac{7}{2.65}\right)=2.64576$$

$$x_3=\frac{1}{2}\left(2.64576+\frac{7}{2.64576}\right)=2.64575$$

$$x_4=\frac{1}{2}\left(2.64575+\frac{7}{2.64575}\right)=2.64575$$

$\therefore \sqrt{7}=2.64575$ متطابقتان لخمس مراتب عشرية x_3, x_4

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1. $x_{n+1}=\frac{1}{2}\left(x_n+\frac{a}{x_n}\right)$

2. $x=\frac{1}{a} \Rightarrow a=\frac{1}{x} \Rightarrow a-\frac{1}{x}=0$

$$\therefore f(x)=a-\frac{1}{x} \Rightarrow f'(x)=\frac{1}{x^2}$$

$$x_{n+1}=x_n-\frac{a-\frac{1}{x_n}}{\frac{1}{x_n^2}}=x_n-\frac{\frac{ax_n-1}{x_n}}{\frac{1}{x_n^2}}=x_n-x_n(ax_n-1)=x_n\left(1-(ax_n-1)\right)=x_n\left(2-ax_n\right)$$

Ex(3):-Find $\frac{1}{7}, x_0 = 0.1$

Solu. $x_{n+1} = x_n(2 - ax_n)$

$x_1 = x_0(2 - 7x_0) = 0.1(2 - 0.7) = 0.13$

$x_2 = 0.13(2 - 7 * 0.13) = 0.13(2 - 0.91) = 0.1417$

$x_3 = 0.14285$

$x_4 = 0.14286$

$x_5 = 0.14286$

$\therefore \frac{1}{7} = 0.14286$

H.W

1. $f(x) = x^3 - 4x^2 + 10, x_0 = -1$

2. Calculate $\sqrt{5}, x_0 = 2$

$f(x) = 2x - 1 - 2\sin x, x_0 = 2$